Experiment - 1

AIM

Design an ER diagram for an Indian e-Commerce Platform with entities Customer, Product, Order, OrderItem, Seller, Category, Payment, delivery, Address. Specify primary keys, Composite and Multi-Valued attributes, weak entities, Specialization of Product, and participation constraints.

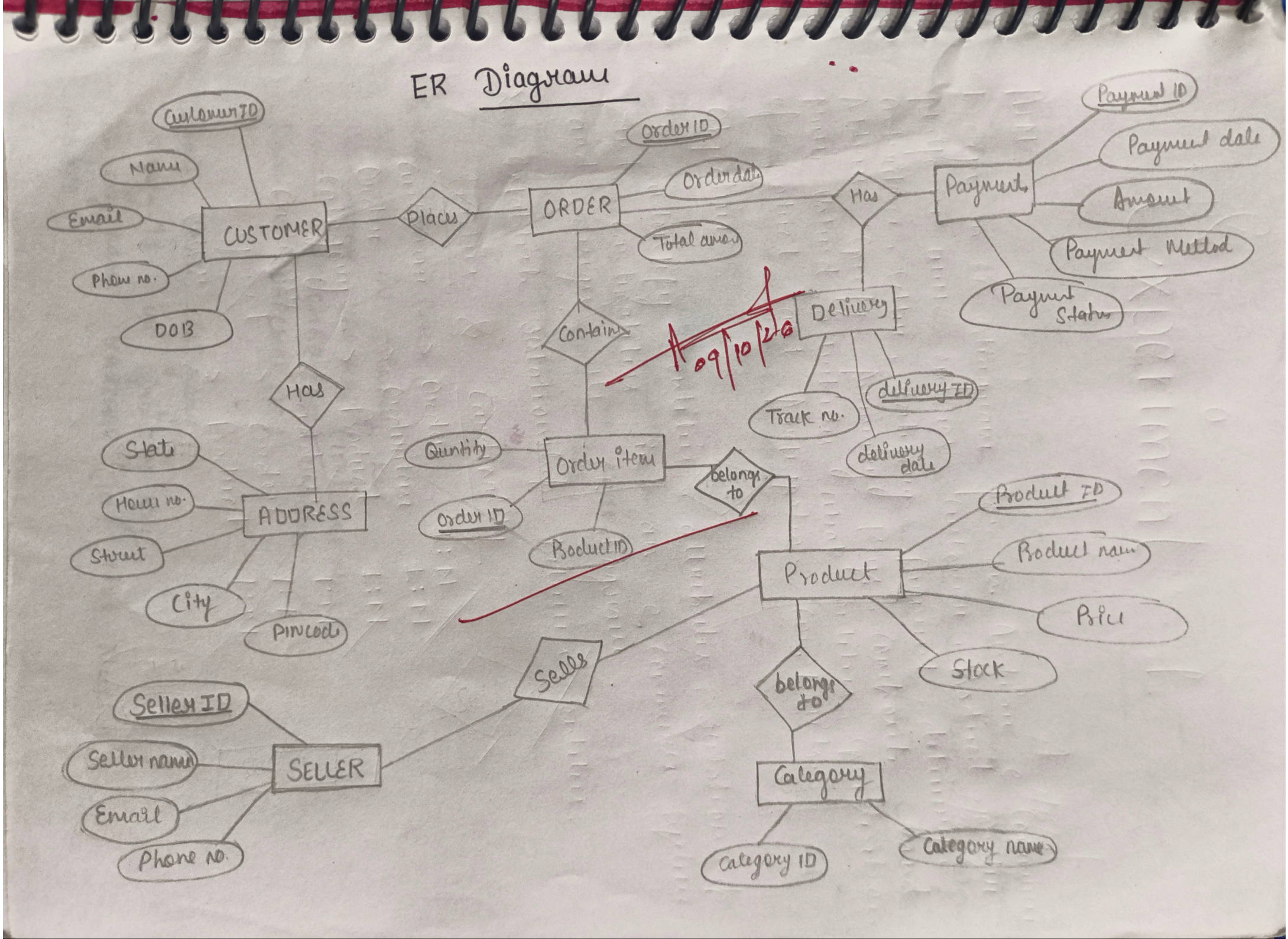
Objective

To identify entities, attributes, relationship, and Cardinalities in an e-Commerce System and design its ER diagram.

※ Establish relationship between entities:

Entities & Cardinality

Customer → Order 1:N
Order → OrderItem 1:N
Category → Product 1:N
Customer → address 1:N
Payment → Order 1:1
Seller → Product 1:N


ER Diagram
CustomerID
Name
Email
CUSTOMER
Phone no.
DOB
Places
ORDER
OrderID
Order date
Total amount
Contain
Has
Payment
Payment ID
Payment date
Amount
Payment Method
Payment Status
Delivery
delivery ID
Track no.
delivery date
09/10/26
Has
State
House no.
Street
City
PINcode
ADDRESS
Quantity
Order item
Order ID
ProductID
belongs to
Product
Product ID
Product name
Price
Stock
Sells
belongs to
SellerID
Seller name
Email
Phone NO.
SELLER
Category
Category ID
Category name